

Exponential Relations over Omnific Integers

Exact toric kernels, decidable fragments,
and one-scale elementary cores

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Abstract

Let $\mathbb{J}$ be the purely infinite surreals, so that Conway’s omnific integers are $\mathbf{Oz} = \mathbb{Z} \oplus \mathbb{J}$. On the additive domain $\mathcal{E}_{\text{alg}} = \overline{\mathbb{Q}} + \mathbb{J} \subseteq \mathbf{No}[i]$, define $\text{Exp}(a + p) = e^a \exp(p)$. Combining Gonshor’s monomial theorem with classical Lindemann–Weierstrass gives an injective exponential group-algebra map. We derive an exact description of all Laurent-polynomial relations among finitely many exponentials, their transcendence degree, and the ordinary complex constants in the field they generate.

The principal model-theoretic application concerns omnific-integer variables and finite exponential sums with algebraic coefficients, algebraic shifts, and real algebraic linear slopes. Their zero relations admit an effective balanced-partition elimination. With addition, order, and congruences, the resulting first-order language is decidable. Its rational-slope reduct is a definitional expansion of Presburger arithmetic. The algebraic-slope structure has a definable decomposition into ordinary Presburger integers and a nonzero ordered vector space over the real algebraic numbers. We classify all its elementary substructures: they are exactly $\mathbb{Z} + W$ for nonzero algebraic-vector subspaces $W \subseteq \mathbb{J}$. In particular, $\mathbb{Z} + \mathbb{A}u$ is a countable elementary core for every positive $u \in \mathbb{J}$.

We also prove a rectangular normal form for definable sets, finite-image results for maps between the two components, and an undecidability boundary obtained by adjoining the single nonlinear relation $\exp(xy) = \exp(z)$. Additional results classify certain exponential equations with arbitrary positive real bases by ordinary integer fibers and purely infinite linear families. With just one irrational scalar-graph predicate, we prove that the theory has exactly the Turing degree of the rational cut of its slope.

Status and scope. An AI-assisted, unrefereed research manuscript. Complete arguments are supplied relative to explicitly stated classical inputs. The main classification, computability, and decidability results are proposed contributions; priority is not certified. No named classical conjecture, full surreal exponential theory, or unrestricted surcomplex exponential problem is claimed solved. Finite verification code is included, but no Lean verification is claimed.

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1 Introduction and principal results

1.1 A restricted language with an exact answer

An omnific integer may contain an arbitrarily complicated set-sized normal form, yet some finite exponential questions about omnific integers see only a surprisingly small part of that complexity. The point of this article is to identify a language in which this observation becomes an exact structural theorem, rather than a heuristic about dominant terms.

Write $\mathbb{A} = \overline{\mathbb{Q}} \cap \mathbb{R}$ for the ordered field of real algebraic numbers. Every $x \in \mathbf{Oz}$ has a unique expression

$$x = n + p, \quad n \in \mathbb{Z}, \quad p \in \mathbb{J}. \quad (1.1)$$

Here “purely infinite” means that every exponent in the Conway normal form is strictly positive, not that the surreal itself is positive. The element p may have either sign, and $p = 0$ is included. In particular, $\mathbb{J}$ is an ordered $\mathbb{R}$ -vector space.

Our exponential predicates have the form

$$R_F(x_1, \dots, x_d) \iff \sum_{j=1}^N c_j \operatorname{Exp}\left(b_j + \sum_{k=1}^d \alpha_{jk} x_k\right) = 0, \quad (1.2)$$

where $c_j, b_j \in \overline{\mathbb{Q}}$ and $\alpha_{jk} \in \mathbb{A}$. Only the variables x_k are quantified, and they range over $\mathbf{Oz}$. The exponential values are not another quantified sort. Products of quantified variables, nested exponentials, and order comparisons between sums of exponentials are not part of the language.

The notation Exp in (1.2) requires no global surcomplex exponential. Its domain is $\overline{\mathbb{Q}} + \mathbb{J}$, and the ordinary complex component is algebraic. The precise construction appears in [section 2](#).

1.2 The main conclusions

Exact algebraic dependence. For any $z_1, \dots, z_d \in \overline{\mathbb{Q}} + \mathbb{J}$, all algebraic relations over $\overline{\mathbb{Q}}$ among $\operatorname{Exp}(z_1), \dots, \operatorname{Exp}(z_d)$ are generated by the integer additive relations among the z_k . In particular,

$$\operatorname{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(\operatorname{Exp}(z_1), \dots, \operatorname{Exp}(z_d)) = \dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}}\{z_1, \dots, z_d\}. \quad (1.3)$$

A stronger statement computes the intersection of this field with $\mathbb{C}$ and establishes linear disjointness over that intersection ([theorems 4.1 and 4.3](#)).

Decidability with algebraic slopes. Let $\mathcal{L}_{\text{alg}}$ be ordered additive arithmetic on $\mathbf{Oz}$, with congruence predicates, expanded by every predicate (1.2). Its theory is decidable, uniformly for effectively specified algebraic coefficients. After a definable coordinate split, every formula becomes a finite union of products of a Presburger-definable integer set and an algebraically semilinear set in $\mathbb{J}$ ([theorems 7.3 and 8.3](#)).

All elementary substructures. A substructure $G \subseteq \mathbf{Oz}$ is elementary for $\mathcal{L}_{\text{alg}}$ if and only if

$$G = \mathbb{Z} + W \quad \text{for a nonzero } \mathbb{A}\text{-vector subspace } W \subseteq \mathbb{J}. \quad (1.4)$$

Consequently, every positive purely infinite u gives a countable elementary substructure $\mathbb{Z} + \mathbb{A}u$. These are additive, relational structures, not exponential subfields or omnific subrings ([theorem 9.1](#) and [corollary 9.2](#)).

A sharp nonlinear obstruction. The same language becomes undecidable after adjoining the single ternary predicate $\exp(xy) = \exp(z)$. This does not depend on an unresolved transcendence conjecture: the new predicate recovers multiplication and the old language already defines the ordinary integers (theorem 11.2).

Exact computational content of one slope. For a language with just the additive structure and the relation $\exp(y) = \exp(\alpha x)$, where α is a fixed ordinary real irrational, its first-order theory has exactly the Turing degree of the rational cut of α . It is decidable exactly for computable slopes (theorem 10.3).

1.3 Relation to existing work and the repository

The Conway normal form, Gonshor exponential, and classical Lindemann–Weierstrass theorem are established inputs, not new claims of this article [1, 2, 3]. Presburger quantifier elimination is also classical [4]. The argument is a combination and application of these inputs. In particular, the linear-independence lemma itself is a short transport argument; the more substantial proposed contribution is the precise language analysis, its definable splitting, and the classification of elementary cores.

The repository VladimirReshetnikov/Surreal was inspected at commit 958b5c4865819bd55ea1f5ffc050282aea7ef570. Its catalogue describes extensive work on omnific Diophantine geometry, set-sized quotients, automorphisms, and initial subgroups [9]. The quotient report explicitly uses $\mathbf{Oz} = \mathbb{Z} \oplus \mathbb{J}$, under its notation for the purely infinite ideal, and studies much stronger ring-theoretic structures [10]. The exponential-automorphism report concerns field endomorphisms compatible with exponentiation, not the additive relational reduct studied here [11].

No advanced theorem from those unrefereed reports is needed in the proofs below. This avoids making our results conditional on their pending review. The overlap is foundational notation and the motivation to distinguish the ordinary constant term from the purely infinite component. A targeted repository search for “Presburger” returned no items but explicitly marked its results incomplete; that is *not* evidence of absence. Targeted public searches also did not identify the precise classification proved here. These bounded searches do not establish priority, and the manuscript makes no exhaustive novelty claim.

2 Normal forms, size conventions, and the exponential domain

2.1 Class-sized objects and finite assertions

We use the standard class of surreal numbers, with each individual surreal a set-sized object and each normal form having set-sized support. Statements about $\mathbf{Oz}$ as a first-order structure are understood formula by formula. Thus $G \preceq \mathbf{Oz}$ means that every fixed finite first-order formula, with a finite tuple of parameters in G , has the same truth value in the two structures. No set containing all surreals, no class-indexed sum, and no unrestricted class-saturation principle is used.

All group algebras and field extensions in the algebraic arguments are formed from set-sized subgroups, usually finitely generated ones. Transcendence degree is used only for the resulting set-sized fields. The proofs involving the entire $\mathbb{J}$ are finite algebraic or first-order elimination arguments and therefore also give schemes of ordinary set-theoretic assertions.

2.2 The purely infinite component

The normal form of a surreal is

$$x = \sum_{\gamma \in S} r_\gamma \omega^\gamma, \quad (2.1)$$

where $S \subseteq \mathbf{No}$ is a set, reverse well ordered, and the nonzero coefficients r_γ are real. Uniqueness, termwise addition, convolution multiplication, and comparison by the leading nonzero term are the facts needed below [1, Chapter 5]; see also [2, Theorem 2.10]. Define

$$\mathbb{J} = \{p \in \mathbf{No} : \text{supp}(p) \subseteq \mathbf{No}_{>0}\}, \quad \mathbf{Oz} = \mathbb{Z} \oplus \mathbb{J}. \quad (2.2)$$

The empty support represents zero. Finite unions of the relevant supports and the usual normal-form multiplication show that $\mathbb{J}$ is an $\mathbb{R}$ -vector space and is closed under multiplication. Consequently, $\mathbf{Oz}$ is a subring of $\mathbf{No}$. The coefficient of ω^0 gives a ring homomorphism $\text{ct} : \mathbf{Oz} \rightarrow \mathbb{Z}$. Put $\text{pp}(x) = x - \text{ct}(x)$.

Lemma 2.1. *Every nonzero $p \in \mathbb{J}$ satisfies $|p| > r$ for every positive real r . In particular, $\mathbb{J} \cap \mathbb{C} = \{0\}$ inside $\mathbf{No}[i]$, and the order on $\mathbf{Oz}$ is lexicographic:*

$$n + p < m + q \iff p < q \vee (p = q \wedge n < m). \quad (2.3)$$

The least positive omnific integer is 1.

Proof. The leading exponent of a nonzero purely infinite normal form is positive. Its leading monomial dominates every positive real, and its sign is that of the leading coefficient. Apply this to $p - q$ to obtain (2.3). If a positive omnific integer has nonzero purely infinite part, it is larger than every positive integer; otherwise it is an ordinary positive integer. The assertion about $\mathbb{C}$ follows since a real surreal that is an ordinary complex number is real. $\square$

Proposition 2.2 (Division with remainder). *For every ordinary integer $m \geq 1$ and $x \in \mathbf{Oz}$, there are unique $y \in \mathbf{Oz}$ and $r \in \{0, 1, \dots, m-1\}$ such that $x = my + r$. Moreover,*

$$x \equiv y \pmod{m} \iff \text{ct}(x) \equiv \text{ct}(y) \pmod{m}. \quad (2.4)$$

Proof. For $x = n + p$, write $n = mq + r$ in $\mathbb{Z}$ and take $y = q + p/m$. The space $\mathbb{J}$ is divisible, so this is an omnific integer. Applying ct to two candidate decompositions gives ordinary uniqueness of r and of the integer quotient; then the purely infinite parts agree as well. The same calculation gives (2.4). $\square$

Together with the ordered abelian group axioms, discreteness and division with remainder are the standard axioms of a Presburger group. Thus $(\mathbf{Oz}, +, -, <, 0, 1, (\equiv_m)_{m \geq 2})$ is a model of the complete theory of the corresponding structure on $\mathbb{Z}$ [4]. Congruence means divisibility in the additive group. The map ct is not order preserving: $\omega - 100$ is positive even though its constant term is negative.

2.3 The two transcendence inputs

Let $\exp : \mathbf{No} \rightarrow \mathbf{No}_{>0}$ denote Gonshor's increasing exponential group isomorphism, extending the real exponential. Write $\mathfrak{M} = \{\omega^\gamma : \gamma \in \mathbf{No}\}$ for the Conway monomial group. We use the established identity

$$\exp(\mathbb{J}) = \mathfrak{M}. \quad (2.5)$$

This is Gonshor's theorem, not the assertion that $\exp(p) = \omega^p$. See [1, Theorems 10.8–10.13], as restated in [2, Theorem 2.16]. Different purely infinite arguments have different monomial exponentials.

The classical Lindemann–Weierstrass theorem, in the form we use, says

$$\begin{aligned} a_1, \dots, a_s \in \overline{\mathbb{Q}} \text{ distinct} &\implies \\ e^{a_1}, \dots, e^{a_s} &\text{ are linearly independent over } \overline{\mathbb{Q}}. \end{aligned} \quad (2.6)$$

The a_j are ordinary complex algebraic numbers. A modern proof of this classical result is given in [3]. Neither (2.5) nor (2.6) is reproved here. All uses of them will be explicit.

Lemma 2.3 (Complex coefficients on real monomials). *Distinct Conway monomials are linearly independent over $\mathbb{C}$ inside $\mathbf{No}[i]$.*

Proof. Separate the real and imaginary parts of a finite purported dependence. Uniqueness of the real Conway normal form makes both coefficients at every monomial zero. $\square$

2.4 A canonical restricted surcomplex exponential

Definition 2.4. Set $\mathcal{E}_{\text{alg}} = \overline{\mathbb{Q}} + \mathbb{J} \subseteq \mathbf{No}[i]$. For its unique decomposition $z = a + p$, define

$$\text{Exp}(z) = e^a \exp(p). \quad (2.7)$$

Uniqueness follows from lemma 2.1. This domain is an additive $\mathbb{Q}$ -vector space. It is not asserted to be a complex algebra or a ring: for example, $i \cdot p$ need not belong to it for nonzero $p \in \mathbb{J}$. Its real part $\mathbb{A} + \mathbb{J}$ is a subring, and on that real domain (2.7) is precisely the restriction of Gonshor's exponential.

The map Exp is a homomorphism from the additive group to $\mathbf{No}[i]^\times$. Only the ordinary complex exponential is used for imaginary parts, and those imaginary parts are algebraic and finite. No sine or cosine at an infinite imaginary argument is chosen. This limited surcomplex extension is sufficient for all coefficients and shifts in (1.2).

3 Faithful exponential group algebras

Theorem 3.1 (Finite exponential independence). *If $z_1, \dots, z_s$ are distinct elements of $\mathcal{E}_{\text{alg}}$, then $\text{Exp}(z_1), \dots, \text{Exp}(z_s)$ are linearly independent over $\overline{\mathbb{Q}}$.*

Proof. Write $z_j = a_j + p_j$ and suppose $\sum_{j=1}^s c_j \text{Exp}(z_j) = 0$, where $c_j \in \overline{\mathbb{Q}}$. Group the finite sum according to the distinct values of p_j . This gives

$$\sum_p \left(\sum_{j:p_j=p} c_j e^{a_j} \right) \exp(p) = 0. \quad (3.1)$$

By (2.5), the outer factors $\exp(p)$ are distinct Conway monomials. Their coefficients are ordinary complex numbers. Hence lemma 2.3 gives $\sum_{j:p_j=p} c_j e^{a_j} = 0$ for every group. Within a fixed group the a_j are distinct, since the original z_j were distinct. Apply (2.6) separately to each group to obtain $c_j = 0$ for every j . $\square$

Remark 3.2. The outer sum in (3.1) is finite. The p_j may individually have arbitrary set-sized reverse-well-ordered supports. No exchange of exponentiation with an infinite sum is made in the proof; all such analytic-normal-form content is confined to the imported theorem (2.5).

Corollary 3.3. *The map $\text{Exp} : \mathcal{E}_{\text{alg}} \rightarrow \mathbf{No}[i]^\times$ is injective. If $z \in \mathcal{E}_{\text{alg}}$ and $\text{Exp}(z) \in \overline{\mathbb{Q}}$, then $z = 0$ and $\text{Exp}(z) = 1$.*

Proof. Equality of two distinct exponential values, or a relation $\text{Exp}(z) - c \text{Exp}(0) = 0$ with $z \neq 0$, contradicts theorem 3.1. $\square$

For a set-sized additive subgroup $H \subseteq \mathcal{E}_{\text{alg}}$, the group algebra $\overline{\mathbb{Q}}[H]$ consists of finite sums $\sum c_h[h]$, with $[h][k] = [h + k]$.

Corollary 3.4 (Faithful group algebra). *The map*

$$\Phi_H : \overline{\mathbb{Q}}[H] \longrightarrow \mathbf{No}[i], \quad \sum_h c_h[h] \longmapsto \sum_h c_h \text{Exp}(h) \quad (3.2)$$

is an injective $\overline{\mathbb{Q}}$ -algebra homomorphism.

Proof. Multiplicativity follows from the additive exponential law. Injectivity is theorem 3.1 after collecting equal group elements. $\square$

Example 3.5. The pair $\omega, \sqrt{2}\omega$ is $\mathbb{Q}$ -linearly independent even though its members lie on one real-linear ray. Consequently their exponentials are algebraically independent, even over $\mathbb{C}$, by the next section. The identity $\exp(\sqrt{2}\omega) = \exp(\omega)^{\sqrt{2}}$ does not contradict this: a fixed irrational real power is not a polynomial operation.

4 Exact toric kernels and ordinary constants

4.1 All finite algebraic relations

Fix $z_1, \dots, z_d \in \mathcal{E}_{\text{alg}}$ and put

$$H = \sum_{k=1}^d \mathbb{Z}z_k, \quad L = \{\ell \in \mathbb{Z}^d : \ell_1 z_1 + \dots + \ell_d z_d = 0\}. \quad (4.1)$$

The group H is finitely generated and torsion free. The relation lattice L is saturated: $m\ell \in L$ for a nonzero integer m implies $\ell \in L$.

Theorem 4.1 (Exact toric kernel). *For the Laurent evaluation map*

$$\text{ev}_z : \overline{\mathbb{Q}}[X_1^{\pm 1}, \dots, X_d^{\pm 1}] \rightarrow \mathbf{No}[i], \quad X_k \longmapsto \text{Exp}(z_k), \quad (4.2)$$

one has

$$\ker(\text{ev}_z) = (X^\ell - 1 : \ell \in L). \quad (4.3)$$

Its image is isomorphic to $\overline{\mathbb{Q}}[H]$. If $r = \dim_{\mathbb{Q}} \text{span}_{\mathbb{Q}}\{z_1, \dots, z_d\}$, then the image is a Laurent polynomial algebra in r variables, and

$$\text{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(\text{Exp}(z_1), \dots, \text{Exp}(z_d)) = r. \quad (4.4)$$

For ordinary polynomial evaluation, the kernel is generated by all $X^u - X^v$ with $u, v \in \mathbb{N}^d$ and $u - v \in L$.

Proof. Send X^u to $[\sum u_k z_k]$. This gives a surjection from the Laurent algebra to $\overline{\mathbb{Q}}[H]$. Composing it with the injective map in [corollary 3.4](#) is exactly ev_z . A Laurent polynomial $\sum c_u X^u$ lies in its kernel if and only if the sum of its coefficients in every coset of L is zero. In each finite coset group, choose one exponent u_0 . The corresponding polynomial is then a linear combination of $X^u - X^{u_0}$, each of which is the Laurent unit X^{u_0} times $X^{u-u_0} - 1$. This proves (4.3).

Choose a $\mathbb{Z}$ -basis of the free group H . Its cardinality is r , and its group algebra is $\overline{\mathbb{Q}}[T_1^{\pm 1}, \dots, T_r^{\pm 1}]$. Taking fraction fields gives (4.4). The same coefficient-grouping argument for a polynomial, using only its nonnegative exponent vectors, proves the final assertion directly. $\square$

Corollary 4.2 (Independence criterion). *A finite family in $\mathcal{E}_{\text{alg}}$ is $\mathbb{Q}$ -linearly independent if and only if its exponentials are algebraically independent over $\overline{\mathbb{Q}}$. In particular, this holds for every finite family of omnific integers.*

A $\mathbb{Z}$ -basis of L suffices to generate the Laurent ideal in (4.3): use $X^{u+v} - 1 = X^u(X^v - 1) + (X^u - 1)$ and the analogous identity for $-u$. This makes the theorem constructive *when a relation lattice is supplied*. It is not an algorithm for discovering all rational relations among arbitrary, uncoded surreal inputs.

4.2 Exactly which standard complex numbers occur

Let

$$\begin{aligned} H_0 &= H \cap \mathbb{C} = H \cap \overline{\mathbb{Q}}, \\ K &= \overline{\mathbb{Q}}(\text{Exp}(h) : h \in H_0), \quad F = \overline{\mathbb{Q}}(\text{Exp}(h) : h \in H). \end{aligned} \tag{4.5}$$

The second description of H_0 follows from $\mathbb{J} \cap \mathbb{C} = 0$.

Theorem 4.3 (Constant field and linear disjointness). *The fields F and $\mathbb{C}$ are linearly disjoint over K , and*

$$F \cap \mathbb{C} = K. \tag{4.6}$$

If s is the $\mathbb{Q}$ -rank of the purely infinite parts of $z_1, \dots, z_d$, then

$$\text{trdeg}_{\mathbb{C}} \mathbb{C}(\text{Exp}(z_1), \dots, \text{Exp}(z_d)) = s. \tag{4.7}$$

Proof. The subgroup H_0 is saturated in H : if a nonzero integer multiple of h has zero purely infinite part, then h has zero purely infinite part. Thus H/H_0 is finitely generated torsion free, hence free, and the exact sequence splits. Choose a basis of H_0 and elements $h_1, \dots, h_s$ representing a basis of a complement. The purely infinite parts $p_1, \dots, p_s$ of these h_j are $\mathbb{Q}$ -linearly independent. Put $t_j = \text{Exp}(h_j)$.

For distinct integer exponent vectors u , the purely infinite elements $\sum u_j p_j$ are distinct. A finite Laurent-polynomial relation in the t_j with complex coefficients therefore becomes a relation among distinct Conway monomials, with each coefficient multiplied by a nonzero ordinary exponential. By [lemma 2.3](#) all coefficients vanish. Hence $t_1, \dots, t_s$ are algebraically independent over $\mathbb{C}$, not merely over K .

We have $F = K(t_1, \dots, t_s)$. The map $\mathbb{C} \otimes_K K[t_1, \dots, t_s] \rightarrow \mathbb{C}[t_1, \dots, t_s]$ is an isomorphism, and localizing at the nonzero polynomials over K stays injective in $\mathbb{C}(t_1, \dots, t_s)$. This is linear disjointness. If $c \in F \cap \mathbb{C}$ but $c \notin K$, the K -independent pair $1, c$ would be linearly dependent over $\mathbb{C}$, a contradiction. The transcendence-degree statement follows from the same algebraic independence and the chosen complement. $\square$

Corollary 4.4 (The omnific constant-field formula). *For $x_1, \dots, x_d \in \mathbf{Oz}$, let $H = \sum \mathbb{Z}x_k$ and write $H \cap \mathbb{Z} = g\mathbb{Z}$ with $g \geq 0$. Then*

$$\overline{\mathbb{Q}}(\exp x_1, \dots, \exp x_d) \cap \mathbb{C} = \begin{cases} \overline{\mathbb{Q}}, & g = 0, \\ \overline{\mathbb{Q}}(e^g), & g > 0. \end{cases} \quad (4.8)$$

Example 4.5 (A hidden ordinary exponential). For $x_1 = \omega$ and $x_2 = 2\omega + 1$, the arguments are $\mathbb{Q}$ -linearly independent, so their exponentials are algebraically independent over $\overline{\mathbb{Q}}$. Nevertheless,

$$\frac{\exp(2\omega + 1)}{(\exp \omega)^2} = e. \quad (4.9)$$

Their generated field is $\overline{\mathbb{Q}}(e, \exp \omega)$ and its intersection with $\mathbb{C}$ is exactly $\overline{\mathbb{Q}}(e)$.

Example 4.6 (An algebraic extension need not be present). For $x_1 = \omega$, $x_2 = \omega + 2$, the corresponding intersection is $\overline{\mathbb{Q}}(e^2)$, not $\overline{\mathbb{Q}}(e)$. Indeed, e is transcendental and $T \notin \overline{\mathbb{Q}}(T^2)$: the automorphism $T \mapsto -T$ fixes the latter field but not T . The existence of a square root in the ambient surcomplex field does not force that root to belong to a specified subfield.

Example 4.7 (Finite coefficients do not automatically become constants). For $x_1 = \omega + 1$ and $x_2 = \sqrt{2}\omega$, their purely infinite parts are $\mathbb{Q}$ -linearly independent. The ordinary complex constants in the generated field are exactly $\overline{\mathbb{Q}}$, even though $\exp(x_1) = e \exp(\omega)$ visibly contains an ordinary transcendental coefficient in its normal form.

5 Balanced partitions eliminate finite exponential equations

Definition 5.1. For coefficients $c = (c_1, \dots, c_N) \in \overline{\mathbb{Q}}^N$, a partition $\mathcal{P}$ of $\{1, \dots, N\}$ is *balanced* if $\sum_{j \in B} c_j = 0$ for every block $B \in \mathcal{P}$. Denote the finite set of balanced partitions by $\text{Bal}(c)$.

Theorem 5.2 (Balanced-partition elimination). *For any $z_1, \dots, z_N \in \mathcal{E}_{\text{alg}}$ and $c_1, \dots, c_N \in \overline{\mathbb{Q}}$,*

$$\sum_{j=1}^N c_j \text{Exp}(z_j) = 0 \iff \bigvee_{\mathcal{P} \in \text{Bal}(c)} \bigwedge_{B \in \mathcal{P}} \bigwedge_{j \in B} (z_j = z_{b(B)}), \quad (5.1)$$

where $b(B)$ is a fixed representative of B . No inequalities between different blocks are required.

Proof. Partition the indices according to their actual equal exponent values. By [theorem 3.1](#), the exponential sum vanishes if and only if the coefficient sum in every actual equality class is zero. This actual partition then occurs on the right of (5.1).

Conversely, if a right-hand conjunction holds, each block contributes zero:

$$\text{Exp}(z_{b(B)}) \sum_{j \in B} c_j = 0.$$

Adding these finitely many zeros gives the left side. Distinct chosen blocks may have equal exponent values; their zero contributions simply coalesce. This explains why no cross-block disequality is needed. $\square$

For $N = 0$ the sum is zero, and the right side is the true empty conjunction associated with the empty partition. A nonempty list with no balanced partition gives false. Zero coefficients may be deleted or allowed as singleton blocks.

Corollary 5.3 (Affine exponential zero loci). *The zero set in $\mathbf{Oz}^d$ of (1.2) is a finite union of sets defined by finite systems of algebraic-affine linear equations. The union and those systems are effectively obtainable from the algebraic coefficients, shifts, and slopes.*

Proof. Every affine exponent lies in $\mathcal{E}_{\text{alg}}$. Apply theorem 5.2, and replace each within-block equality by

$$\sum_k (\alpha_{jk} - \alpha_{b(B)k})x_k + (b_j - b_{b(B)}) = 0. \quad (5.2)$$

Algebraic addition and equality decide balance exactly. No exponential constant has to be approximated numerically. $\square$

Remark 5.4 (An effective transcendence application). Classical Lindemann–Weierstrass justifies a symbolic identity test. The algorithm does not search for a numerical lower bound separating nearly equal exponential sums. Even with ordinary complex exponentials, zero testing reduces to algebraic coefficient sums and affine exponent equality.

5.1 Worked equations

Proposition 5.5. *For omnific-integer variables the following equivalences hold:*

$$\exp x + \exp y = 2 \iff x = y = 0; \quad (5.3)$$

$$\exp x + \exp y = \exp u + \exp v \iff (x = u \wedge y = v) \vee (x = v \wedge y = u); \quad (5.4)$$

$$\exp(x + y) - \exp x - \exp y + 1 = 0 \iff x = 0 \vee y = 0. \quad (5.5)$$

More generally, equality of two finite sums of unit-coefficient exponentials is equivalent to equality of the multisets of their exponents, including multiplicities.

Proof. For (5.3), the coefficient list $(1, 1, -2)$ has no proper nonempty zero-sum subset; all three exponents $x, y, 0$ must agree. For the multiset assertion, group equal exponents and apply theorem 3.1; the net integer multiplicity in each group is zero. This gives (5.4). Finally, the left side of (5.5) is $(\exp x - 1)(\exp y - 1)$ in a field. The factors vanish exactly at zero by injectivity. Alternatively theorem 5.2 yields the same alternatives. $\square$

The conclusion in (5.3) is special to the prescribed exponent domain; it is false over all real arguments. A positive real $t \neq 1$ with $t < 2$ gives real exponents $\log t, \log(2 - t)$. Such logarithmic constants need not belong to $\mathbb{A} + \mathbb{J}$.

6 The languages and the rational-slope theorem

6.1 Formal syntax and effective input

The base language is

$$\mathcal{L}_{\text{Pr}} = \{+, -, <, 0, 1, (\equiv_m)_{m \geq 2}\}. \quad (6.1)$$

All its function symbols act on the sole domain $\mathbf{Oz}$. The language $\mathcal{L}_{\text{alg}}$ adds one relation symbol R_F for each finite specification (1.2); its coefficients and shifts are ordinary complex algebraic numbers and its slopes are real algebraic numbers. The sublanguage $\mathcal{L}_{\text{rat}}$ uses only rational slopes, while retaining algebraic coefficients and algebraic shifts.

An effective algebraic-number code consists, for example, of a defining rational polynomial and a root-isolating interval or complex rectangle. Standard algebraic number arithmetic decides validity, equality, field operations, realness, and the sign of real algebraic numbers. A finite family can be embedded effectively in one number field and expressed in a rational basis. These are the algebraic computations used by the decision procedure. They can equivalently be included as an exact algebraic-number subroutine specification; standard algorithms are treated in [6].

Ordinary integer parameters are terms in 1. Arbitrary surreal constants are not part of the effective input alphabet. Semantic results with arbitrary finite parameter tuples will be stated separately; they do not imply an algorithm for reading arbitrary surreal normal forms.

6.2 Reduction to Presburger arithmetic

Lemma 6.1. *For rational $q_1, \dots, q_d$ and algebraic $b \in \overline{\mathbb{Q}}$, the relation*

$$q_1x_1 + \dots + q_dx_d + b = 0, \quad x_k \in \mathbf{Oz}, \quad (6.2)$$

is either empty or is definable by one integer-affine equation in $\mathcal{L}_{Pr}$. This distinction and the equation are effective.

Proof. The rational combination of the x_k belongs to $\mathbb{Q} + \mathbb{J}$. If it equals the ordinary algebraic number $-b$, its purely infinite part vanishes and b is rational. Thus for nonrational b there are no solutions. If b is rational, clear all denominators. The resulting equation uses only integer multiples, addition, and an ordinary integer constant, which are terms of $\mathcal{L}_{Pr}$. $\square$

Theorem 6.2 (Rational slopes add no Presburger expressive power). *Every $\mathcal{L}_{rat}$ -formula translates effectively to an equivalent $\mathcal{L}_{Pr}$ -formula on $\mathbf{Oz}$. Consequently its theory is decidable and*

$$(\mathbb{Z}, \mathcal{L}_{rat}) \preceq (\mathbf{Oz}, \mathcal{L}_{rat}), \quad (6.3)$$

where the predicates on $\mathbb{Z}$ are obtained by the same exponential interpretation.

Proof. Use [theorem 5.2](#) to eliminate every exponential predicate and [lemma 6.1](#) on each resulting affine equality. Leave the Boolean connectives and quantifiers unchanged. This is a formula-by-formula translation, so it applies to arbitrary alternations of quantifiers, not just existential sentences.

The base structures on $\mathbb{Z}$ and $\mathbf{Oz}$ satisfy the complete Presburger theory. The inclusion of $\mathbb{Z}$ preserves its quantifier-free language, including congruences by [\(2.4\)](#). Presburger quantifier elimination therefore makes this inclusion elementary. Applying the translation proves [\(6.3\)](#). The same elimination decides sentences effectively [\[4\]](#). $\square$

Remark 6.3. The theory contains order *on the omnific variables*, not order comparisons between exponential sums. Nor does the use of a rational slope declare division to be a total function on $\mathbf{Oz}$. Rational multiplication is used only in the ambient affine exponent, and the final relations are expressed after clearing denominators.

7 Algebraic slopes and effective two-sort elimination

7.1 Splitting every linear equation

Lemma 7.1 (Algebraic coordinate splitting). *Let $\alpha_k \in \mathbb{A}$ and $b \in \overline{\mathbb{Q}}$. Write $x_k = n_k + p_k$ as in (1.1). Then*

$$\sum_k \alpha_k x_k + b = 0 \iff \left(\sum_k \alpha_k p_k = 0 \right) \wedge \left(\sum_k \alpha_k n_k + b = 0 \right). \quad (7.1)$$

The second conjunct is effectively equivalent to a finite system of integer-affine equations in the n_k .

Proof. The first sum on the right is in $\mathbb{J}$; the second is an ordinary complex number. Their intersection is zero, so the split is exact. Let B be a number field containing the finitely many α_k and b , and choose a rational basis $\beta_1, \dots, \beta_t$ of B . Write $\alpha_k = \sum_h a_{hk} \beta_h$ and $b = \sum_h b_h \beta_h$ with rational coordinates. The ordinary equation is equivalent to $\sum_k a_{hk} n_k + b_h = 0$ for every h . Clear denominators in each equation. $\square$

Coefficients in a number-field basis are *fixed rational numbers*; they are not new quantified variables. This step introduces no multiplication of integer variables.

7.2 Quantifier elimination for the vector component

Let $\mathcal{L}_{\mathbb{A}\text{-vec}}$ be the language of ordered abelian groups with a unary scalar map λ_α for every $\alpha \in \mathbb{A}$. The intended structures are nonzero ordered $\mathbb{A}$ -vector spaces. No scalar-field sort is quantified.

Lemma 7.2 (Ordered vector-space elimination). *The theory of nonzero ordered $\mathbb{A}$ -vector spaces has quantifier elimination and is complete and decidable. Every order-preserving $\mathbb{A}$ -linear embedding between such spaces is elementary.*

Proof. All terms are affine linear expressions in vector variables with fixed algebraic scalar coefficients and, when parameters are allowed, fixed vector parameters. To eliminate one existential variable, put the quantifier-free formula into a finite disjunction of conjunctions. Replace a disequality by its two strict-order alternatives, and a weak inequality by its equality and strict-order alternatives. It is enough to eliminate y from a conjunction of linear equalities and strict inequalities.

If an equality has a nonzero coefficient of y , solve it uniquely for y and substitute in the other conditions. If no equality has such a coefficient, retain the equalities independent of y . After dividing by nonzero coefficients and reversing inequalities when their coefficients are negative, every remaining condition involving y is $a_i < y$ or $y < b_j$. Existence is equivalent to all $a_i < b_j$ when both kinds of bound occur. For finitely many such bounds, take the maximum lower bound and minimum upper bound; their midpoint is a witness. With only lower bounds or only upper bounds, unboundedness gives a witness. With no bounds, nonemptiness suffices. Ordered nonzero vector spaces are dense and have no endpoints, since halving a positive difference and adding a positive vector are available.

This proves effective elimination of one variable. Induction eliminates all quantifiers. Scalar arithmetic, equality, and signs in $\mathbb{A}$ are decidable. A parameter-free quantifier-free sentence has only the vector constant zero, so its truth is independent of the particular nonzero ordered vector space. This gives completeness and decidability. Finally, a linear order embedding preserves every quantifier-free formula, hence every formula by elimination. $\square$

This elementary proof is the linear case of the general ordered-vector-space setting studied in the literature on linear o-minimal structures [5]. It also explains why different vector-space dimensions or Archimedean ranks need not be distinguished by first-order sentences in this language: scalars are named operations, not quantified elements of a field sort.

7.3 The translation theorem

Theorem 7.3 (Effective two-sort reduction). *Every $\mathcal{L}_{\text{alg}}$ -formula on $\mathbf{Oz}$ translates effectively, under $x_k = n_k + p_k$, into a formula in the disjoint two-sorted structure*

$$(\mathbb{Z}, \mathcal{L}_{\text{Pr}}) \sqcup (\mathbb{J}, \mathcal{L}_{\mathbb{A}\text{-vec}}). \quad (7.2)$$

The theory of $(\mathbf{Oz}, \mathcal{L}_{\text{alg}})$ is decidable. The translated formulas admit effective quantifier elimination in the two-sorted language.

Proof. First replace each exponential atom by [theorem 5.2](#). Apply [lemma 7.1](#) to the resulting algebraic-affine equalities. This yields conjunctions of Presburger equations in n and vector equations in p .

For base-language atoms, addition and negation split coordinatewise, constants $0, 1$ become $(0, 0), (1, 0)$, congruence is given by [\(2.4\)](#), and order is given by [\(2.3\)](#). For example,

$$x_1 + 2x_2 < 3 \iff p_1 + 2p_2 < 0 \vee (p_1 + 2p_2 = 0 \wedge n_1 + 2n_2 < 3). \quad (7.3)$$

Equality splits into equality of both coordinates. Replace an existential quantifier over x by $\exists n \in \mathbb{Z} \exists p \in \mathbb{J}$, and a universal quantifier by the corresponding pair of universal quantifiers. The bijection [\(1.1\)](#) proves the translation correct by induction on formulas.

Every atomic formula of the result belongs to exactly one of the two sorts. Finite Boolean combinations can be put into a finite disjunction of formulas $A(n) \wedge B(p)$. To eliminate an integer variable, distribute its existential quantifier over this disjunction and use Presburger elimination in A ; B does not involve that variable. Eliminate a vector variable in the same way using [lemma 7.2](#). Universal quantifiers are handled by negation. This terminates for any fixed finite input formula. For a sentence, the final truth value is effectively determined. No bound on computational efficiency is asserted. $\square$

Example 7.4 (An irrational equation separates the coordinates). The relation

$$\text{Exp}(y) = \text{Exp}(\sqrt{2}x), \quad x, y \in \mathbf{Oz}, \quad (7.4)$$

is equivalent to $y = \sqrt{2}x$. Its coordinate conditions are $p_y = \sqrt{2}p_x$ and $n_y - \sqrt{2}n_x = 0$. Expanding in the rational basis $1, \sqrt{2}$ forces $n_y = n_x = 0$. Thus [\(7.4\)](#) describes a scalar map on purely infinite elements and cannot have a nonzero ordinary integer input. This observation is the key to the next section.

8 Definable coordinates and a normal form for all definable sets

8.1 Recovering the integer and purely infinite sorts internally

Theorem 8.1 (Definable splitting). *In $(\mathbf{Oz}, \mathcal{L}_{\text{alg}})$ the subsets $\mathbb{J}$ and $\mathbb{Z}$, and the graph of ct , are parameter-free definable. Explicit definitions are*

$$\text{Pure}(x) \quad :\Longleftrightarrow \quad \exists y (\text{Exp}(y) = \text{Exp}(\sqrt{2}x)), \quad (8.1)$$

$$\text{Int}(x) \quad :\Longleftrightarrow \quad \forall p (\text{Pure}(p) \wedge p > 0 \implies -p < x < p), \quad (8.2)$$

$$\text{CT}(x, n) \quad :\Longleftrightarrow \quad \text{Int}(n) \wedge \text{Pure}(x - n). \quad (8.3)$$

For every $\alpha \in \mathbb{A}$, the scalar action on $\mathbb{J}$ is definable by

$$\text{Pure}(p) \wedge \text{Pure}(q) \wedge (\text{Exp}(q) = \text{Exp}(\alpha p)). \quad (8.4)$$

Consequently the structure is definably equivalent to the two-sorted structure (7.2), with reconstruction map $(n, p) \mapsto n + p$.

Proof. By injectivity, the witness in (8.1) must be $y = \sqrt{2}x$. For $x = n + p$, this is an omnific integer exactly when $\sqrt{2}n \in \mathbb{Z}$, which happens exactly for $n = 0$. Thus (8.1) defines $\mathbb{J}$.

An ordinary integer is strictly between $-p$ and p for every positive $p \in \mathbb{J}$, by lemma 2.1. Conversely, if $x = n + p_0$ and $p_0 \neq 0$, take $p = |p_0|/2 \in \mathbb{J}_{>0}$. The leading purely infinite comparison gives $|x| > p$, so (8.2) fails. Hence it defines exactly $\mathbb{Z}$. The unique decomposition (1.1) now proves (8.3). Equation (8.4) is the graph $q = \alpha p$, again by injectivity.

The induced ordered additive structure on the integer sort is Presburger arithmetic, and on the pure sort we have all ordered $\mathbb{A}$ -vector operations. Theorem 7.3 defines every original relation from these two sorts. Conversely (8.1)–(8.4) and the inherited group operations define the two sorts and their coordinate projections in the original structure. The forward and reverse constructions compose via the definable bijection $(n, p) \mapsto n + p$. This proves the stated equivalence, including the definable identification of each composition with the original structure. $\square$

Remark 8.2. Definability of ordinary $\mathbb{Z}$ does not by itself imply undecidability. Here its induced structure contains addition and order but not multiplication. The explicit reduction proves that no hidden integer multiplication is supplied by the exponential-equality predicates.

8.2 Rectangular normal form and orthogonality

Call a subset of $\mathbb{J}^d$ $\mathbb{A}$ -semilinear if it is a finite Boolean combination of affine linear equalities and inequalities with scalar coefficients in $\mathbb{A}$ and fixed vector parameters in $\mathbb{J}$. Equivalently, it is a finite union of sets specified by finite systems of such conditions.

Theorem 8.3 (Rectangular normal form). *Every $\mathcal{L}_{\text{alg}}$ -definable set $X \subseteq \mathbf{Oz}^d$, allowing a finite tuple of parameters, has a description*

$$X = \bigcup_{j=1}^r \{n + p : n \in A_j \subseteq \mathbb{Z}^d, p \in B_j \subseteq \mathbb{J}^d\}, \quad (8.5)$$

where A_j is Presburger definable and B_j is $\mathbb{A}$ -semilinear. The parameters needed in the A_j are ordinary integer coordinates of the original parameters; the vector parameters needed in the B_j are their purely infinite coordinates. Conversely every set of the form (8.5) is $\mathcal{L}_{\text{alg}}$ -definable.

Proof. Treat the original parameters as additional free variables during the translation in [theorem 7.3](#). Quantifier elimination in the two sorts gives a finite Boolean combination of sort-separated atoms. Put it into disjunctive normal form; each conjunction is the intersection of one integer condition and one vector condition. Presburger elimination and [lemma 7.2](#) give the indicated forms. This proves (8.5) and the parameter statement. The converse uses the definable coordinate maps and scalar graphs from [theorem 8.1](#). $\square$

Corollary 8.4 (No extra induced structure). *The integer and purely infinite subsets are stably embedded: a definable subset of either of their finite Cartesian powers can be defined using parameters from that same subset. Their induced structures are exactly Presburger arithmetic and ordered $\mathbb{A}$ -vector-space geometry, respectively.*

Proof. In (8.5), a subset of $\mathbb{Z}^d$ is obtained by setting $p = 0$. Each B_j either contains zero or does not, leaving a finite union of the corresponding A_j . Similarly set $n = 0$ for a subset of $\mathbb{J}^d$. The converse inclusions of the induced languages were proved in [theorem 8.1](#). $\square$

Corollary 8.5 (Finite-image cross-component maps). *Every definable map from a definable subset of $\mathbb{Z}^m$ to $\mathbb{J}^n$ has finite image. The same holds for a definable map from a subset of $\mathbb{J}^m$ to $\mathbb{Z}^n$, even with parameters from $\mathbf{Oz}$.*

Proof. The graph is a finite union of rectangles in the two sorts. A nonempty rectangle $A \times B$ contained in a function graph can have at most one element in its output factor B . Thus each nonempty rectangle contributes at most one output. The proof in the reverse direction is identical. $\square$

This is an exact separation statement, not merely a dimension comparison. For example, there is no definable injection of the infinite set $\mathbb{Z}$ into $\mathbb{J}$ in this language, despite the existence of many set-theoretic injections and the ambient operation $n \mapsto n\omega$. That operation would require multiplication of a variable by the nonalgebraic parameter ω , which is not one of the allowed scalar maps on omnific variables.

9 Elementary substructures, embeddings, and one-scale cores

Theorem 9.1 (Classification of elementary substructures). *For a substructure $G \subseteq (\mathbf{Oz}, \mathcal{L}_{\text{alg}})$ the following are equivalent:*

- (i) $G \preccurlyeq (\mathbf{Oz}, \mathcal{L}_{\text{alg}})$;
- (ii) $G = \mathbb{Z} + W$ for a nonzero $\mathbb{A}$ -vector subspace $W \subseteq \mathbb{J}$, equipped with its inherited order and all inherited exponential-equality predicates.

The assertion applies to set-sized G , and formula by formula to class-sized substructures as well.

Proof. Suppose first that G is elementary. It is an additive subgroup containing 1, so it contains every ordinary integer. If $x \in G$, its actual integer constant term $n = \text{ct}(x)$ is therefore in G , and $x - n \in G \cap \mathbb{J}$. Thus $G = \mathbb{Z} + (G \cap \mathbb{J})$. Set $W = G \cap \mathbb{J}$.

The sentence $\exists p (\text{Pure}(p) \wedge p > 0)$ holds in $\mathbf{Oz}$, so by elementarity it holds in G . Since Pure is definable and elementarity preserves its truth on parameters in G , this gives a nonzero member of W . For $p \in W$ and fixed $\alpha \in \mathbb{A}$, the sentence with parameter p asserting existence of a pure q satisfying (8.4) is true in $\mathbf{Oz}$. It is therefore true in G . Its witness must equal the actual αp

by injectivity of Exp . Hence W is closed under every algebraic scalar, and is a nonzero $\mathbb{A}$ -vector subspace.

Conversely, let $G = \mathbb{Z} + W$ with $W \neq 0$ an $\mathbb{A}$ -vector subspace. It is a substructure for the group operations and constants, with the relations inherited from $\mathbf{Oz}$. The coordinate translation of [theorem 7.3](#) works in G with the two sorts $\mathbb{Z}$ and W : real algebraic combinations of the pure coordinates remain in W , and lexicographic order and congruence have the same formulas. By [lemma 7.2](#), $W \preceq \mathbb{J}$ in the vector language. The identity on the integer sort is elementary. Therefore every translated formula with parameters from $\mathbb{Z} \sqcup W$ has the same truth value in $\mathbb{Z} \sqcup W$ and $\mathbb{Z} \sqcup \mathbb{J}$. Translating back gives $G \preceq \mathbf{Oz}$. $\square$

Corollary 9.2 (One-scale elementary cores). *For every $u \in \mathbb{J}_{>0}$,*

$$G_u = \{n + au : n \in \mathbb{Z}, a \in \mathbb{A}\} \preceq (\mathbf{Oz}, \mathcal{L}_{\text{alg}}). \quad (9.1)$$

The set G_u is countable. For positive $u, v \in \mathbb{J}$, the map $n + au \mapsto n + av$ is an isomorphism of these structures.

Proof. The map $a \mapsto au$ identifies the nonzero ordered $\mathbb{A}$ -vector space $\mathbb{A}$ with $\mathbb{A}u$. Apply [theorem 9.1](#). Countability follows from that of $\mathbb{Z}$ and $\mathbb{A}$. The displayed map is the identity on the integer coordinates and an ordered $\mathbb{A}$ -linear isomorphism on the pure coordinates, so the coordinate translation preserves every relation and formula. $\square$

For $u = \omega$, the set G_ω is not closed under multiplication: $\omega^2 \notin \mathbb{Z} + \mathbb{A}\omega$ by normal-form uniqueness. It is also not asserted to contain its exponential values. Those values are used externally to interpret relations, just as a relation on integers may be defined using a larger field. This is why (9.1) is compatible with the immense field-theoretic complexity of the full surreal numbers.

Corollary 9.3 (Classification of elementary embeddings). *Let $G_W = \mathbb{Z} + W$ and $G_V = \mathbb{Z} + V$ be elementary substructures as above. Their elementary embeddings are exactly the maps*

$$n + p \mapsto n + T(p), \quad (9.2)$$

where $T : W \rightarrow V$ is an order-preserving $\mathbb{A}$ -linear embedding. In particular,

$$\text{Aut}(G_W, \mathcal{L}_{\text{alg}}) \cong \text{Aut}_{\mathbb{A}, <}(W). \quad (9.3)$$

Proof. An elementary embedding fixes ordinary integers, preserves the definable pure subset and the graph of ct , and preserves every scalar graph (8.4). Its restriction to W is therefore an order-preserving $\mathbb{A}$ -linear embedding, and additivity gives (9.2). Conversely, [lemma 7.2](#) makes every such T elementary on vector coordinates. Together with the identity on $\mathbb{Z}$, the two-sort translation proves elementarity of (9.2). The automorphism statement is the bijective case. $\square$

Proposition 9.4 (The two slope languages genuinely differ). *The inclusion $\mathbb{Z} \subseteq \mathbf{Oz}$ is elementary for $\mathcal{L}_{\text{rat}}$ but not for $\mathcal{L}_{\text{alg}}$.*

Proof. The first assertion is [theorem 6.2](#). In $\mathcal{L}_{\text{alg}}$, the sentence $\exists x (x > 0 \wedge \text{Pure}(x))$ is true in $\mathbf{Oz}$. In the induced structure on $\mathbb{Z}$, the defining equation $y = \sqrt{2}x$ has only the solution $x = y = 0$, so the sentence is false. $\square$

Remark 9.5 (Why one scale can be elementary). A formula has only finitely many fixed algebraic scalar coefficients. It can specify finitely many linear bounds and equations, but it cannot quantify over all scalars to say that one positive vector is larger than every scalar multiple of another. The vector-space elimination proof shows exactly why a nonempty interval left by finitely many constraints already contains a witness in an ordered algebraic-linear subspace. The “one scale” assertion is a first-order-language phenomenon, not a claim that all surreal growth rates collapse arithmetically.

10 One irrational slope and an exact computability classification

The preceding arguments also identify the precise computational content of a much smaller language. Here the slope may be an arbitrary ordinary real irrational, not necessarily algebraic. This extension admits only its scalar graph, not all exponential sums with that slope.

Definition 10.1. For a fixed $\alpha \in \mathbb{R} \setminus \mathbb{Q}$, let $\mathcal{S}_\alpha$ be the structure on $\mathbf{Oz}$ in the finite signature $\{+, -, <, 0, 1, S\}$, where

$$S(x, y) \iff \exp(y) = \exp(\alpha x). \quad (10.1)$$

Put $K_\alpha = \mathbb{Q}(\alpha)$, as an ordered subfield of $\mathbb{R}$. Congruence predicates can be added definitionally by $x \equiv_m y \iff \exists z (x - y = mz)$.

The real argument αx is a legitimate surreal argument of Gonshor’s exponential even when its ordinary constant part is transcendental. Real exponential injectivity gives $S(x, y) \iff y = \alpha x$. No use of Lindemann–Weierstrass is needed in this section.

Theorem 10.2 (Single-slope decomposition). *The structure $\mathcal{S}_\alpha$ is definably equivalent to the disjoint union of Presburger $\mathbb{Z}$ and the nonzero ordered K_α -vector space $\mathbb{J}$. Its elementary substructures are exactly*

$$\mathbb{Z} + W, \quad 0 \neq W \subseteq \mathbb{J} \text{ a } K_\alpha\text{-vector subspace.} \quad (10.2)$$

In particular, $\mathbb{Z} + K_\alpha u$ is a countable elementary core for every $u \in \mathbb{J}_{>0}$.

Proof. Write $x = n + p$, $y = m + q$. Since α is irrational,

$$S(x, y) \iff n = m = 0 \wedge q = \alpha p. \quad (10.3)$$

Indeed, the ordinary component equation $m = \alpha n$ forces $m = n = 0$. Thus $\exists y S(x, y)$ defines $\mathbb{J}$, and the formula (8.2), with this new pure-part definition, defines $\mathbb{Z}$. The decomposition and constant-term graph are then definable as before.

On $\mathbb{J}$ the relation S is the graph of scalar multiplication by α . Composition and addition give every polynomial in this operator with rational coefficients. Rational scalar maps are definable by divisibility in the vector group. For a nonzero polynomial value $g(\alpha)$, the graph of multiplication by $f(\alpha)/g(\alpha)$ is defined by the equation $g(\alpha)q = f(\alpha)p$ using these polynomial operator graphs. It has exactly one solution q for every p . Hence every scalar map of K_α is definable.

Conversely, the two coordinate sorts define S by (10.3) and the base group language by lexicographic order and coordinatewise addition. The ordered-vector-space elimination proof of lemma 7.2 works verbatim over any fixed ordered scalar field; effectiveness is not needed for this semantic claim. The same arguments as in theorem 9.1 therefore yield (10.2). The field $\mathbb{Q}(\alpha)$ is countable, which proves the last assertion. $\square$

For the computational statement, encode rational numbers and integer polynomials by natural numbers. Let

$$D_\alpha = \{q \in \mathbb{Q} : q < \alpha\}, \quad P_\alpha = \{f \in \mathbb{Z}[T] : f(\alpha) > 0\}. \quad (10.4)$$

For sets of finite codes, $A \leq_T B$ means that an algorithm with a membership oracle for B decides membership in A ; $\equiv_T$ means reductions both ways.

Theorem 10.3 (The theory has the degree of the slope). *For every fixed ordinary real irrational α ,*

$$\text{Th}(\mathcal{S}_\alpha) \equiv_T P_\alpha \equiv_T D_\alpha. \quad (10.5)$$

In particular, $\text{Th}(\mathcal{S}_\alpha)$ is decidable if and only if α is a computable real number. The reductions are statements about each fixed α ; the final equivalence does not assert uniform zero testing from arbitrary names of computable reals.

Proof. First, an oracle for P_α decides the signs and zero values of polynomials at α : zero is equivalent to neither f nor $-f$ belonging to P_α . It therefore performs exact ordered-field arithmetic with rational-function representatives in K_α . Translate a sentence using (10.3) and eliminate its integer and vector quantifiers as before, consulting this oracle for scalar equality, division, and sign. This proves $\text{Th}(\mathcal{S}_\alpha) \leq_T P_\alpha$.

In the reverse direction, write $f(T) = \sum_{j=0}^d a_j T^j$. Construct a sentence asserting that there are $p_0, \dots, p_d \in \mathbb{J}$ with $p_0 > 0$, $S(p_j, p_{j+1})$ for $j < d$, and $\sum_j a_j p_j > 0$. Here $\mathbb{J}$ is replaced by its defining formula, and integer multiples are additive terms. The recurrence forces $p_j = \alpha^j p_0$, so the sentence holds exactly when $f(\alpha) > 0$. This is an effective reduction $P_\alpha \leq_T \text{Th}(\mathcal{S}_\alpha)$.

The reduction $D_\alpha \leq_T P_\alpha$ uses the linear polynomials $nT - m$ for $q = m/n$ and $n > 0$. To prove the converse for a fixed α , distinguish two cases. If α is algebraic, polynomial signs at it are decidable by exact algebraic arithmetic, with an algebraic representation of this fixed number built into the algorithm. If α is transcendental, every nonzero integer polynomial has a nonzero value at α . The cut oracle provides nested rational intervals containing α with arbitrarily small widths. Evaluate f by rational interval arithmetic until its value interval excludes zero. This terminates for every nonzero f ; the zero polynomial is handled separately. Thus $P_\alpha \leq_T D_\alpha$ in both cases.

Finally, for an irrational real α , its rational cut is decidable precisely when it is computable. A decidable cut gives rational approximations by bisection. Conversely, computable approximations decide comparison with any fixed rational by refinement, since equality with a rational is impossible. This completes the proof. $\square$

The countable elementary core in [theorem 10.2](#) also shows directly that the first-order truth set in (10.5) is the theory of a set-sized structure. There is no appeal to a global satisfaction class for a proper-class universe.

Proposition 10.4 (Nonuniformity of arbitrary real-number names). *There is no single algorithm that, given a program computing rational approximations with prescribed error bounds to an algebraic irrational α , and a sentence of the signature of $\mathcal{S}_\alpha$, always decides that sentence. This remains false even when every supplied name is valid and every slope is promised to be algebraic irrational.*

Proof. For a machine index e , define

$$\alpha_e = \begin{cases} \sqrt{2} + 2^{-t}, & \text{if machine } e \text{ first halts at step } t, \\ \sqrt{2}, & \text{if it never halts.} \end{cases} \quad (10.6)$$

There is a uniformly computable approximation name for α_e : simulate the machine sufficiently many steps for the requested precision; if it has not halted, the effect of any later halting is at most the corresponding power of 2^{-1} . Combine this bound with a rational approximation to $\sqrt{2}$. Every α_e is algebraic irrational, irrespective of whether the machine halts.

A fixed sentence of the scalar-graph language expresses $\alpha_e^2 > 2$: choose pure $p > 0$, apply S twice to obtain $q = \alpha_e^2 p$, and require $q > 2p$. The sentence is true exactly when the machine halts. A uniform decision procedure from those approximation names would therefore decide the halting problem, which is impossible. The argument does not supply a minimal polynomial or an isolating-root representation of α_e uniformly; those are the stronger algebraic codes required by our earlier positive algorithm. $\square$

This gives a second boundary, independent of nonlinear exponentiation. Even a single partial scalar graph can carry noncomputable information when its real slope is not effectively known. For algebraic slopes with the explicit algebraic codes specified in [section 6](#), this obstruction is absent.

11 Boundaries: tails, constants, and a single undecidable extension

11.1 Allowing arbitrary finite constants

The algebraicity assumption on finite exponent constants is essential to [theorem 3.1](#). Allowing $\log 2$ immediately gives

$$\exp(\log 2) - 2 \exp(0) = 0. \quad (11.1)$$

The arguments are distinct. This does not contradict Lindemann–Weierstrass: $\log 2$ is not algebraic, since a nonzero algebraic argument cannot have the algebraic exponential 2. The failure is located at an explicitly excluded finite constant.

11.2 Allowing infinitesimal tails

Proposition 11.1 (A tail obstruction). *Finite exponential independence over $\overline{\mathbb{Q}}$ fails on the full surreal field even when the coefficients are integers.*

Proof. Take $\varepsilon = \omega^{-1}$, $a = \log(1 + \varepsilon)$, and $b = \log \varepsilon$. Then $a > 0$, $b < 0$, and both differ from zero. The exponential inverse law in **No** gives

$$\exp(a) - \exp(b) - \exp(0) = (1 + \varepsilon) - \varepsilon - 1 = 0. \quad (11.2)$$

The element a is a nonzero infinitesimal, so it does not lie in $\mathbb{A} + \mathbb{J}$. Indeed, for every positive real r , one has $1 + \varepsilon < e^r$, so $0 < a < r$ by monotonicity. By [\(2.5\)](#), b , as the logarithm of a monomial, is purely infinite. Thus one infinitesimal-tail argument already produces a nontrivial three-term relation. $\square$

The proposition does not claim that $\mathcal{E}_{\text{alg}}$ is maximal among all additive domains with finite exponential independence. Maximality is a separate extension problem; see [research question 3](#).

11.3 Nonlinear exponents recover multiplication

Theorem 11.2 (A single nonlinear relation destroys decidability). *Expand $(\mathbf{Oz}, \mathcal{L}_{\text{alg}})$ by the ternary predicate*

$$M(x, y, z) \iff \exp(xy) = \exp(z). \quad (11.3)$$

The first-order theory of the expansion is undecidable.

Proof. The product xy is an omnific integer because $\mathbf{Oz}$ is a ring. Injectivity of the real exponential makes (11.3) equivalent to $xy = z$. Thus the new relation is the graph of multiplication on $\mathbf{Oz}$.

The old formula (8.2) defines exactly the ordinary integers. Relativize all quantifiers in an ordinary arithmetic sentence to Int , replace each multiplication term by its graph using M , and retain ordinary addition, order, 0, and 1. This is an effective interpretation of the true first-order arithmetic of $\mathbb{Z}$, equivalently of $\mathbb{N}$ after restricting to nonnegative integers. That theory is undecidable by the classical incompleteness and undecidability results for arithmetic [7, 8]. A decision procedure for the expansion would decide it, a contradiction. $\square$

This is a statement about the *full first-order theory*. The relativization uses a quantified definition of $\mathbb{Z}$; no claim about an existential-only reduction is being silently made. Likewise, this argument is not a claim that every nonlinear exponential predicate leads to undecidability.

11.4 What the decision theorem does not cover

The language analyzed here does not contain multiplication of two variable omnific integers, the graph of $x \mapsto \exp x$ as an internal function, arbitrary surreal scalar multiplication, arbitrary real or complex coefficient constants, or exponential inequalities. It also has no complex-variable quantifiers and no exponential at an infinitely large imaginary argument.

In particular, allowing an inequality $\alpha x < y$ with an irrational algebraic α on the *finite integer coordinates* is a different extension: it creates integer cuts involving irrational slopes rather than the rational linear equalities produced in lemma 7.1. The current proof does not establish a decision procedure for that extension. It would be incorrect to infer one merely because algebraic slopes are allowed inside *equality* predicates.

12 Beyond algebraic coefficients: exact fibers for real-base equations

There is a useful extension of the grouping argument that does not require Lindemann–Weierstrass. It gives geometry over the purely infinite coordinates but leaves a genuinely ordinary exponential equation on the integer coordinates.

Theorem 12.1 (Ordinary-fiber criterion). *Let $c_j \in \mathbb{C}$, $\lambda_{jk} \in \mathbb{R}$, and consider*

$$F(x) = \sum_{j=1}^N c_j \exp\left(\sum_k \lambda_{jk} x_k\right), \quad x \in \mathbf{Oz}^d. \quad (12.1)$$

For a fixed $n \in \mathbb{Z}^d$, put

$$d_j(n) = c_j e^{\sum_k \lambda_{jk} n_k}, \quad P_j(p) = \sum_k \lambda_{jk} p_k. \quad (12.2)$$

Then $F(n+p) = 0$ if and only if there exists a partition $\mathcal{P}$ of $\{1, \dots, N\}$ such that for every block B ,

$$\sum_{j \in B} d_j(n) = 0, \quad P_j(p) = P_{b(B)}(p) \quad (j \in B). \quad (12.3)$$

Thus every fixed integer fiber is a finite union of real-linear subspaces of $\mathbb{J}^d$. No uniform decision procedure for which partitions satisfy the ordinary coefficient equations is asserted.

Proof. Rewrite (12.1) as $\sum_j d_j(n) \exp(P_j(p))$. Since each $P_j(p) \in \mathbb{J}$, distinct values produce distinct Conway monomials by (2.5). Their complex coefficients vanish separately by lemma 2.3. The balanced-partition argument now applies to these fixed complex numbers $d_j(n)$ without any use of their algebraicity. Within-block equality is real-linear and homogeneous in p , which gives the fiber description. $\square$

Corollary 12.2 (Positive real-base ternary equations). *Fix ordinary real numbers $a, b, c > 1$. All solutions in $\mathbf{Oz}^3$ to*

$$a^x + b^y = c^z, \quad a^x := \exp((\log a)x), \quad (12.4)$$

are exactly

$$x = n + \frac{\lambda}{\log a}, \quad y = m + \frac{\lambda}{\log b}, \quad z = k + \frac{\lambda}{\log c}, \quad (12.5)$$

where $n, m, k \in \mathbb{Z}$ satisfy $a^n + b^m = c^k$ and $\lambda \in \mathbb{J}$ is arbitrary.

Proof. In a fixed integer fiber the three nonzero coefficients have signs $+, +, -$. A partition into at least two blocks would have a block consisting only of positive coefficients or the lone negative coefficient, so it cannot be balanced. Hence all three purely infinite exponents agree: $(\log a) \text{pp}(x) = (\log b) \text{pp}(y) = (\log c) \text{pp}(z) = \lambda$. Their common coefficient equation is the ordinary equation $a^n + b^m - c^k = 0$. This proves necessity and gives (12.5). Substitution proves sufficiency. Division by the fixed nonzero real logarithms preserves $\mathbb{J}$. $\square$

Example 12.3. Since $2^1 + 3^1 = 5^1$, every $\lambda \in \mathbb{J}$ gives the solution

$$2^{1+\lambda/\log 2} + 3^{1+\lambda/\log 3} = 5^{1+\lambda/\log 5}. \quad (12.6)$$

Each exponent is an omnific integer. For $\lambda > 0$ they are all positive infinite. Their ordinary constant terms are nevertheless exactly 1.

Corollary 12.4 (An inhomogeneous anchor removes all infinite solutions). *For fixed ordinary $a, b > 1$, every omnific-integer solution of*

$$a^x - b^y = 1 \quad (12.7)$$

has $x, y \in \mathbb{Z}$. Thus (12.7) has exactly its ordinary integer solutions.

Proof. The three coefficients in each integer fiber have signs $+, -, -$, so again only the one-block partition can be balanced. The third purely infinite exponent is zero because it represents the constant 1. Therefore $(\log a) \text{pp}(x) = (\log b) \text{pp}(y) = 0$, whence both purely infinite parts vanish. $\square$

These are transfer and parameterization theorems. They do not solve the ordinary integer equations left on the base. In the homogeneous equation (12.4), positivity of an infinite omnific variable does not force its ordinary constant term to be positive, so restrictions on n, m, k must not be inferred from order conditions on x, y, z without using (2.3).

13 Effective procedures and finite verification

13.1 A complete mathematical decision pipeline

The proofs provide the following terminating algorithm for a finite $\mathcal{L}_{\text{alg}}$ sentence. This is a mathematical specification of the full procedure, not a claim that the accompanying small program implements all its stages.

1. Normalize every relation atom to a finite exponential sum. Products of exponential factors with allowed affine exponents can be flattened using $\text{Exp}(u)\text{Exp}(v) = \text{Exp}(u + v)$; Laurent powers also cause no difficulty. Remove zero coefficients when convenient.
2. Enumerate the set partitions of the finite list of terms. Keep those with zero algebraic coefficient sum in every block. For each retained partition, output within-block affine exponent equalities.
3. Introduce integer and purely infinite coordinates for every omnific variable. Use [lemma 7.1](#) and rational-basis arithmetic in the finite number fields occurring in the input. Translate order and congruence by [\(2.3\)](#) and [\(2.4\)](#).
4. Eliminate quantifiers in the two disjoint sorts, using Presburger elimination and the ordered-vector-space procedure in [lemma 7.2](#). Evaluate the resulting closed Boolean expression.

This deliberately simple partition algorithm may produce B_N branches, where B_N is the N th Bell number, before accounting for later Boolean and Presburger expansion. No polynomial-time or elementary-time complexity bound is claimed.

The toric-kernel calculation is a different interface: it starts with a finite argument family together with an exact description of its additive relation lattice. Given that lattice, basis reduction supplies finitely many Laurent binomial generators. Arbitrary surreal input equality is not solved by treating a finite coefficient specification as if it were a full normal-form oracle.

13.2 What the supplied program checks

The file `code/checks.py` uses exact rational arithmetic and exact arithmetic in $\mathbb{Q}(\sqrt{2})$. It contains a reusable finite balanced-partition generator and checks the following layers independently of numerical exponential evaluation:

1. balanced partitions versus direct coefficient grouping for exhaustive small coefficient and exponent-label lists, including empty and zero-coefficient cases;
2. the finite-support forms of [\(5.3\)](#), [\(5.4\)](#), and [\(5.5\)](#);
3. the $\sqrt{2}$ pure-part detector and the coefficientwise splitting of algebraic linear equations, using exact quadratic-field coordinates;
4. the inhomogeneous and homogeneous fiber criterion with positive and negative ordinary coefficients at representative formal scales.

The recorded run passed 22,074 finite cases, including 7,381 exhaustive partition cases, 6,561 two-term multiset cases, and 2,000 quadratic-coordinate splitting cases. The run is deterministic, uses no network access, and writes its actual counts and outcomes to `check_results.json`. Exponential values in these checks are represented in a finite formal group algebra. Therefore the tests check the

combinatorial and coordinate implementations, not Gonshor’s theorem or Lindemann–Weierstrass. They do not formally verify all quantifiers or all proper-class assertions of the article.

13.3 A formalization path

A useful formalization starts with an abstract additive group E , a field K , and an injective group-algebra evaluation into a field extension. The toric and partition theorems then have no surreal prerequisites. A second layer formalizes an ordered algebraic-vector space V and the lexicographic group $\mathbb{Z} \oplus V$, with finite algebraic-linear relation predicates. The splitting, rectangular normal form, and elementary-substructure arguments belong to this layer. Only the final bridge needs the concrete surreal normal form and Gonshor exponential. This separation mirrors the distinction in the repository between generic checked algebra and its concrete surreal applications [9, 11].

14 Further research questions

The questions below are proposed continuations of this manuscript. They are not presented as a catalogue of previously named open problems, and no priority claim is attached to their formulation.

Research question 1 (Exponential inequalities). Which expansions by signs or order comparisons of finite exponential sums remain decidable? A first target is rational slopes with real algebraic coefficients. One must account for the leading monomial and the sign of its ordinary exponential coefficient, not merely its vanishing. With algebraic irrational slopes, clarify exactly which induced integer cuts appear.

Research question 2 (Countably many scalar operators). [Section 10](#) classifies the structure with one irrational scalar graph. For an effectively indexed countable family of named real slopes containing an irrational slope, determine the oracle degree of the resulting theory in terms of the ordered field they generate. Distinguish uniform approximation of the slopes from uniform knowledge of all finite polynomial relations among them. For finite families, also specify which extra presentation data make the reductions uniform. The nonuniformity in [proposition 10.4](#) shows why a list of approximation programs is not automatically enough.

Research question 3 (Extensions of the independent exponential domain). Classify additive domains $E \supseteq \overline{\mathbb{Q}} + \mathbb{J}$ on which an extension of the exponential group-algebra map over $\overline{\mathbb{Q}}$ remains injective. The counterexample (11.2) shows that arbitrary infinitesimal tails cannot all be admitted. It does not exclude carefully chosen transcendental constants or special tail subspaces. A useful objective is a criterion in terms of the monomial and unit components of their exponentials; the extension of the exponential itself must also be specified.

Research question 4 (More general ordinary coefficient fields). For an effectively presented field of ordinary constants larger than $\overline{\mathbb{Q}}$, identify sufficient independence and zero-testing hypotheses for the same partition reduction. Distinguish unconditional transfers from those requiring new ordinary transcendence information. Arbitrary real-base equations in [section 12](#) provide a test case where only the purely infinite geometry is automatic.

Research question 5 (The nonlinear boundary below multiplication). The single ternary predicate in [theorem 11.2](#) recovers multiplication. Determine which smaller nonlinear families avoid this. Unary nonlinear predicates, restricted compositions, or predicates whose nonlinearity is confined to a known pure subspace are concrete candidates. Every proposal must be tested against definable recovery of integer multiplication.

Research question 6 (Parameter-uniform effective theories). Develop a computable presentation of selected parameter families in $\mathbb{J}$, including an algorithm for the finite algebraic-linear order relations that the vector-space elimination procedure asks. Determine when the elementary diagram of such a parameter expansion is decidable. The arbitrary-parameter semantic theorem in [theorem 8.3](#) alone is not such an algorithm.

Research question 7 (Complexity and compact certificates). Replace exhaustive balanced-partition enumeration by zero-sum circuit methods, matroid data, or shared Boolean representations. Give upper and lower complexity bounds for fixed arity, fixed coefficient field, and bounded numbers of exponential terms. A certificate should record both the coefficient balance and the affine equalities, so that it is independently checkable.

Research question 8 (Toric and arithmetic families). Study uniform families of the constant-field theorem as the additive relation lattice changes. Which ordinary constant fields can occur over prescribed pure parts? How do finite-index changes such as $\overline{\mathbb{Q}}(e^2)$ versus $\overline{\mathbb{Q}}(e)$ interact with specialization and normalization? The exact intersection formula is a starting point, not a classification of all larger exponential subfields.

Research question 9 (Model-theoretic refinements). Use the definable product description to study types, definable groups, canonical parameters, and elementary embeddings in countable models of the resulting complete theory. Relate the classification in [theorem 9.1](#) to the repository's much stronger ring- and field-embedding questions without confusing their languages. In particular, characterize additional predicates that first make a one-scale core non-elementary.

Research question 10 (A larger canonical surcomplex domain). Find useful, explicitly specified domains permitting more imaginary components while retaining a canonical exponential and a controlled kernel. The present domain avoids all infinite imaginary phases. A broader construction must state its phase convention and track period relations before an exponential independence or decidability theorem can be formulated correctly.

Research question 11 (Machine-checked bridge). Formalize the abstract group-algebra and coordinate-splitting theorems first, then supply the concrete bridge from Conway normal forms to $\exp(\mathbb{J}) = \mathfrak{M}$. Record exactly which parts are proved about an abstract model and which are instantiated for the actual surreal field. This is a realistic route to checking the entire dependency chain rather than only finite examples.

15 Conclusion

The exponential relations studied here have an exact algebraic explanation: monomial separation handles the purely infinite component, and Lindemann–Weierstrass handles the algebraic ordinary component. Every finite relation then reduces to balanced additive coincidences of its exponents.

The resulting first-order theory is stronger than Presburger arithmetic when algebraic irrational slopes are named, but it still has a precise decidable structure: ordinary additive integers and a nonzero ordered algebraic-vector space, with no additional interaction after definable splitting. This yields the rectangular description of definable sets and the full classification of its elementary substructures. The countable one-scale cores are consequences of that classification, not approximations to the full surreal field. In the single-irrational-slope language, the same splitting identifies the exact Turing degree of the theory with the degree of the slope, while distinguishing fixed-slope decidability from uniform decision using arbitrary approximation names.

The nonlinear extension and the explicit tail counterexample locate real limits of the results. The appropriate conclusion is therefore not a resolution of unrestricted surreal exponentiation, but a proof-bearing structural and algorithmic theory of a substantial, precisely bounded exponential-equality fragment. Independent mathematical review and a broader priority search remain necessary before treating the proposed contributions as established new results.

A Proof dependencies and audit checklist

A.1 Dependencies

The dependency chain is deliberately short. Normal-form uniqueness and $\exp(\mathbb{J}) = \mathfrak{M}$, together with ordinary Lindemann–Weierstrass, imply [theorem 3.1](#). That theorem implies the exact toric kernel and the balanced-partition criterion. The constant-field theorem needs only monomial independence over $\mathbb{C}$ once the pure-part complement is chosen. Balanced partitions and algebraic coordinate splitting reduce the logical language to Presburger arithmetic and ordered-vector-space linear geometry. The latter reduction yields decidability, definability of the two components, rectangular normal form, and the elementary-core classification. The single-slope decomposition uses only exponential injectivity and the same coordinate geometry, not Lindemann–Weierstrass. Its degree calculation is proved by explicit oracle reductions. Classical undecidability of integer arithmetic and of halting is used only for the two negative boundary results.

No conjectural assertion in the source repository is an input. Nor are Berarducci–Mantova differentiation, saturation, strong summability, set-sized quotient rigidity, or a global complex exponential needed.

A.2 Checks against likely overstatements

Potential misreading	Actual assertion and safeguard
$\exp(p) = \omega^p$	Not used. Only the set equality $\exp(\mathbb{J}) = \mathfrak{M}$ and injectivity are imported from Gonshor.
A new classical transcendence theorem	Not claimed. Classical Lindemann–Weierstrass is an explicit input; the finite independence argument is its monomial transport.
A global surcomplex exponential	Not defined. Exp acts only on $\overline{\mathbb{Q}} + \mathbb{J}$, with ordinary finite algebraic imaginary part.
A decision procedure for arbitrary surreal input	Not claimed. The effective input is a finite formula with algebraic coefficient codes. Arbitrary surreal parameters require additional effective data.
Full exponential-field decidability	Not claimed. Only externally interpreted finite-sum zero predicates with affine exponents are admitted.
All algebraic-slope inequalities	Not admitted. Base order compares additive omnific terms; the algebraic slopes occur inside exponential equalities.
Quantifier elimination in the original one-sort signature	Not asserted. The proved elimination is after a definable two-sort coordinate expansion.
Uniform decisions from arbitrary real approximation names	Not claimed. The fixed-slope Turing reductions need not be uniform in such names; proposition 10.4 proves the obstruction.

Potential misreading	Actual assertion and safeguard
A one-scale surreal subfield	Not claimed. $\mathbb{Z} + \mathbb{A}u$ is an additive relational substructure; it is generally not a subring.
Existential undecidability of the nonlinear extension	Not claimed. The proof interprets full first-order arithmetic using a quantified integer definition.
Verified novelty or formal proof	Neither is claimed. The literature search is bounded, the repository search was incomplete, and finite tests are not a Lean formalization.

B Reproducibility and source record

The accompanying archive contains this source, its compiled PDF, `code/checks.py`, its recorded JSON output, a source-and-claim audit, and build instructions. Build with `latexmk -pdf article.tex`; the bibliography is contained in this file, so no external `.bib` file is needed. The checks use Python’s standard library only.

The public source inspection was performed on 23 September 2026. In the Mantova–Matusinski PDF, the normal-form statement and the exponential monomial statement were visually checked on PDF pages 7 and 11, respectively; the latter is Theorem 2.16. In Cluckers’ preprint, the Presburger language and the recalled quantifier-elimination and decidability statements were visually checked on PDF page 2. Popescu’s version 2 abstract states exactly the complex Lindemann–Weierstrass input used here. These checks establish what the references say, not the novelty of the deductions made from them.

The repository catalogue, the exponential-automorphism report overview, and relevant introductory portions of the quotient report were inspected through the GitHub connector at the pinned commit. Its full collection is large; this was a focused comparison of relevant topics, not a claim-by-claim audit of every repository manuscript. The bibliography points to immutable commit URLs for the inspected repository sources.

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